A Three levels of an information-processing system (Marr, 1982)

The same table describes a brain or a machine. Dashed arrows: agreement on one row does not fill the rows below.

	Brain *biological substrate*	**Machine** *artificial substrate*
1 Computational theory *this paper: function* What is computed, and why that mapping is appropriate	Example task: infer an object's location. Specify the behavioral goal.	Example task: predict the next token. Specify the training objective.
loosely coupled		
2 Representation and algorithm *this paper: dynamics* Which internal variables, and how they are transformed	Candidate codes and update mechanisms. Test them against neural measurements.	Tokens, embeddings and attention. Training updates differ from inference.
loosely coupled		
3 Hardware implementation *this paper: architecture* How representation and algorithm are realized physically	Neurons, synapses, spikes. Energy, and a body.	Processors, memory and arithmetic. Measure the physical implementation.

Not a Marr level

Bodily regulation, experience and values raise additional questions. This context band is not a fourth Marr level or a test that settles machine consciousness.